Civic and Discivic Technologies

A Method for Classifying Technologies by What They Do to the Humans Who Use Them

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Abstract

Technologies are often evaluated by what they can do — their features, their capabilities, their convenience. This paper argues for a different evaluation: technologies should be classified by what they do to the humans who use them over time. Some technologies expand human capacity. The human who uses them ends up with more ability than they started with, and the ability persists whether or not the technology is present. These technologies are civic — they build the civilization that uses them. Other technologies substitute for human capacity. The human who uses them ends up with less ability than they started with, or with ability that only works as long as the technology is available. These technologies are discivic — they erode the civilization that depends on them. Most technologies are one or the other by their nature, and some can be either depending on how they are used. The paper develops the distinction, provides a test for applying it, and works through examples ranging from hammers and writing to social media and artificial intelligence. The goal is to give readers a stable way to evaluate technologies in their own lives and in their societies — not by what the technologies promise, but by what the technologies actually produce in the humans who use them.

1. Why You Are Reading This

You use technologies every day. Some of them you chose deliberately — the car you drive, the software you work with, the tools you picked for a specific purpose. Others arrived without any decision on your part — the platforms your work requires, the interfaces your services use, the devices that came with the infrastructure of modern life. All of them are doing something to you, whether or not you tracked it.

Some of what they're doing is useful. They're helping you accomplish things you wanted to accomplish. They're extending your reach, saving your time, connecting you to people and information you couldn't otherwise access. That's the case for adoption — the value you get justifies the use.

But that's not the whole accounting. Technologies also shape what you become as you use them. A technology that requires you to develop skill shapes you into someone with that skill. A technology that handles the skill for you shapes you into someone who doesn't have it. Both technologies might accomplish the same immediate task. The humans they produce, over years of use, are different.

This is the accounting the paper is about. Not whether a technology works — most of them work, in the sense of delivering their advertised function. Whether the technology, across the pattern of use, produces a richer human or a poorer one. Whether your children, growing up inside the technology, will be more capable than you were, or less. Whether the civilization that adopts the technology at scale ends up stronger or weaker for having adopted it.

These are not new questions. Every generation of humans has asked them about the technologies they inherited, adopted, or refused. What the paper contributes is specific vocabulary — civic and discivic — and a specific test, so the questions can be asked precisely rather than as vague unease. With the vocabulary and the test, you can classify the technologies in your own life, explain the classifications to others, and make decisions about adoption, refusal, and child-rearing that are grounded in what the technologies actually do rather than in what they're marketed as.

The paper is not prescriptive about which technologies to use or avoid. That decision depends on your circumstances, your tradeoffs, your values. What the paper provides is the frame for making the decision on honest terms. With the frame, you can still choose a discivic technology if the trade-off serves you — but you'll know it's a trade-off, and you'll know what you're paying for what you're getting. Without the frame, you make the same trade and you don't see the payment, and the payment accumulates anyway.

2. What Technology Does

A technology is a thing humans build to extend what humans can do. That definition is broad on purpose. It covers the first sharpened stone, the plow, the printing press, the telephone, the personal computer, and the latest artificial intelligence. All of them are human constructions deployed to let humans do something that wouldn't be possible or practical without them.

The extending happens in two different ways, and the difference is the subject of this paper.

The first way is augmentation. The technology multiplies a capacity the human already has. A hammer multiplies the force the hand can deliver. A pencil multiplies the marks the finger can make in the dirt. A telescope multiplies what the eye can see. In each case, the human's existing capacity is taken as the input, and the technology produces a larger output than the human could produce alone. The human is still doing the thing. They're doing it better, farther, faster, with more precision. But they're doing it.

The second way is substitution. The technology does the thing instead of the human. An automatic transmission shifts gears so the driver doesn't have to. A calculator performs arithmetic so the student doesn't have to. A navigation system decides the route so the driver doesn't have to. In each case, the technology takes over a function the human could perform, and the human's relationship to the function changes — they're no longer doing it, they're receiving its output.

Both patterns are real, and both can be useful in specific situations. An automatic transmission in traffic, a calculator for a spreadsheet, a GPS in an unfamiliar city — these are cases where substitution serves the person using the technology. They're accomplishing something without having to do the full task. That's a benefit.

The question the paper is about is what happens over time. A human who uses augmentation technology regularly ends up with more capacity than they started with — the augmentation has been building the skill all along, because the human has been doing the thing with the tool's help. A human who uses substitution technology regularly ends up with less capacity — the substitution has been preventing the skill from developing, because the human has been receiving output rather than producing it.

This is not a minor distinction. Over a lifetime, it's the difference between a carpenter who has spent forty years with hand tools and knows more about wood than most people will ever know, and a person who has spent forty years pushing a button that orders precut boards and has no relationship to wood beyond the ordering. Over a generation, it's the difference between children who grow up writing by hand and develop the hand-eye-mind coordination handwriting builds, and children who grow up tapping on glass and develop only the narrow skill of glass-tapping. Over civilizational time, it's the difference between a people who carry their capacities inside them and can rebuild from scratch if their technologies fail, and a people whose capacities live only in the technologies and collapse when the technologies do.

3. The Core Distinction

A civic technology is one that, under regular use, leaves the human user with more capacity than they started with. The capacity is real — it lives in the person, works when the technology is absent, transfers to other situations, and persists across time. The technology has been building the human while the human has been using the technology.

A discivic technology is one that, under regular use, leaves the human user with less capacity than they started with, or with capacity that only works while the technology is available. The person has been receiving the technology's output rather than producing their own, and the capacity that would have been built by producing has not been built.

The distinction is not about how sophisticated the technology is. Ancient technologies can be civic or discivic. Modern technologies can be civic or discivic. The distinction is not about whether the user benefits in the short term — both kinds of technology can help the user accomplish things they wanted to accomplish. The distinction is about what the technology produces in the user over the long term of regular use.

This matters because civilizations are built out of what their people can do. A civilization of carpenters, farmers, writers, readers, thinkers, navigators, craftsmen, problem-solvers, argument-makers, and attentive observers is capable of many things. The things it's capable of are the sum of what its people can do. When those capacities are present in the population, the civilization has resources to draw on, people to call on, skills to deploy when circumstances demand them.

A civilization whose people have less of these capacities, because the technologies they grew up with substituted for them, is a different civilization. It can still do some things — usually the things the technologies can do. It cannot easily do the things the technologies don't cover, because the people don't have the underlying skills. When the technologies fail, or drift, or become unavailable for any reason, the civilization discovers what it cannot do, and the discovery arrives too late to rebuild.

Civic technologies add to civilization. They leave behind more capable humans, more durable skills, a population that can do things in diverse conditions. Discivic technologies subtract from civilization. They leave behind less capable humans, skills that exist only in the infrastructure, a population whose functioning depends on the infrastructure continuing to work.

The stakes of this distinction are not primarily about individual users. Individuals can make trade-offs that suit their circumstances. An adult who grew up writing by hand can safely use autocomplete most of the time — they already have the skill, and autocomplete is handling routine cases while their skill remains available for the cases that matter. The individual is using a mixed technology in a way that works for them.

The stakes are at the population level and across generations. A technology that the current adult population uses without serious harm may, when children grow up inside it from the formation window onward, produce adults who never built the skills the current adults have. Those adults cannot make the same trade-offs the current adults made, because they never developed the underlying capacity the trade-off assumed. The technology that was useful for adults becomes disabling for the next generation, and the loss is not recoverable within those lives.

This is why the classification matters. It's not about whether you, personally, should use a given technology. It's about what the technology, deployed at scale in developmental environments, produces in the humans who form inside it. Some technologies pass this test. Some fail it. Civilizations need to know the difference.

4. The Test

For any technology, ask one question: what remains in the human after regular use?

If the answer is "more capacity — skill, knowledge, judgment, or ability — that works whether or not the technology is present" — the technology is civic. The user has been built by the using.

If the answer is "less capacity — the user can do less on their own than they could before, or less than they would have been able to do if they hadn't used the technology regularly" — the technology is discivic. The user has been diminished by the using.

The test has three features that matter.

First, it looks at the human, not the output. A technology can produce impressive output while leaving the user unchanged or diminished. A novel-writing assistant can produce novels while producing no novelists. A GPS can produce arrivals at destinations while producing no navigators. The output is real; the human-building that would have accompanied genuine production is absent.

Second, it looks at regular use, not occasional use. Almost any technology can be used occasionally without lasting effect on the user. A calculator used once in a lifetime doesn't change the user's arithmetic capacity. A GPS used on one trip through a foreign country doesn't erode the user's navigation skills. The question is what happens when the technology becomes part of daily life — a regular input to the user's functioning, experienced repeatedly across years.

Third, it looks at what persists when the technology is absent. Real capacity works without the tool that helped build it. The carpenter can still plane a board if the planer breaks, because the skill is in the carpenter. The writer can still compose a sentence if the word processor crashes, because the composition is in the writer. Capacity that only works while the tool is running isn't capacity — it's dependency. A person who can only write with autocomplete doesn't have writing capacity; they have autocomplete-assisted-production capacity, which evaporates when autocomplete does.

With the test, you can classify technologies in your own life. Pick one you use regularly. Ask what you can do now that you couldn't do before you adopted it. Ask what you could do before that you can't do as well now. Ask what would happen to your ability to function if the technology became unavailable tomorrow. The answers place the technology somewhere on the civic-discivic spectrum for you, in your pattern of use.

The test is transferable. You can apply it to technologies other people are using, and to technologies you're considering introducing to your children. You can apply it to technologies your society is adopting at scale. In each case, the question is the same: what does this do to the humans who use it over time?

5. The Inner Dimension

Technology classification would be simpler if civic and discivic were properties of the technology itself, independent of how it was used. Some technologies come close to this. A hammer is civic across nearly any pattern of use — it's hard to imagine using a hammer regularly without becoming more capable with hammers. A short-form video platform is discivic across nearly any pattern of use — the platform's design specifically extracts attention while training the user to fragment further.

But many technologies are in between. The same artifact, used differently by different humans, produces different outcomes. A computer used to learn programming, write documents, and create art is a civic technology for that user. The same computer used to passively consume algorithmic feeds and receive automated output is discivic for the user who uses it that way. The artifact is the same. The humans are different, because the relationship to the artifact is different.

What distinguishes the civic from the discivic use of a mixed technology is whether the human stays in the driver's seat. Stays engaged with the work, makes the decisions, produces rather than receives, holds the meaning of what they're doing rather than accepting the tool's output as a substitute for understanding.

A person using a writing tool who types their own thoughts and uses the tool for formatting, searching, and distribution is in the driver's seat. The writing is theirs. The tool is amplifying their production. They're becoming a more capable writer as they work.

A person using the same writing tool who accepts its grammar suggestions, its style corrections, its phrasing improvements, and its auto-completed sentences is not in the driver's seat. The writing is co-produced with the tool, with the tool's judgment shaping the output at every step. They're becoming a more capable director-of-writing-tools, which is a different skill, but they're not becoming a more capable writer.

A person using an artificial intelligence to help with research, where they ask questions and direct follow-ups and verify outputs and make editorial decisions about what matters, is in the driver's seat. They're building skill in research direction and, as they work, absorbing the subject matter they're directing. The AI is augmenting their capacity.

A person using the same AI by delegating the task — asking it to produce a finished output with minimal direction and accepting what it produces — is not in the driver's seat. The AI is producing; the human is receiving. The skill that would have built through direction isn't building. The subject matter that would have been absorbed through directed engagement isn't being absorbed. The same technology is operating in discivic mode for this user.

This is why the classification of many technologies depends on how they're used, and why adults can sometimes use discivic-patterned technologies in civic ways by maintaining active engagement. It's also why children are at particular risk: children don't have the prerequisite skills to maintain the driver's seat position. They don't know what questions to ask. They can't evaluate the tool's output critically. They accept what the tool produces because they have no basis for rejecting it. A technology that an adult can use in civic mode may operate in discivic mode for every child who touches it, because children can't hold the position the civic mode requires.

The design of a technology matters because design determines how easy it is to stay in the driver's seat, and how easy it is to drift into passive reception. A chat interface with an AI keeps the human typing, reading, deciding at each turn — it structurally requires the driver's seat. An agentic system that runs with a single prompt and produces finished output structurally enables delegation — the user can step away and return only to receive. Same underlying capability; different positions the user is placed in. The chat interface leans civic; the agent leans discivic, unless specifically configured otherwise.

Design choices also matter at the population scale. A technology deployed with defaults that enable driver's-seat use across most users will produce more civic outcomes than a technology deployed with defaults that encourage delegation. Providers who choose the delegation default are choosing — sometimes knowingly — to tilt the population toward the discivic outcome. The individual user may be able to fight the default, but most users don't, and the aggregate result follows the default rather than the resisters.

6. Worked Examples

The distinction becomes concrete when applied to specific technologies. The examples below are not exhaustive — readers can add and refine for their own situations. The purpose is to show how the test lands on representative cases, including hard cases where reasonable people disagree.

Writing. Writing is civic. The person who writes regularly develops sustained attention, vocabulary, sentence structure, argument construction, and the ability to produce language from their own mind. The writing itself builds the writer. This holds whether the writing is on paper, on a typewriter, or in a text editor. The specific medium matters less than the activity — the human is composing, phrasing, committing to words. Over a lifetime of writing, the person becomes someone who can think in written language, which is a capacity of general use that extends far beyond writing itself.

Handwriting and cursive in particular. Civic, and in ways that modern schooling has undervalued. Handwriting builds hand-eye-mind coordination that typing does not build. It engages motor control, fine muscle development, and neurological channels that benefit the person across many unrelated tasks. Cursive specifically allows sustained handwritten thought at length — block letters fatigue the hand too quickly for long-form composition. Children who learn cursive can write long by hand; children who don't, can't. The removal of cursive from school curricula has produced children who can type but cannot write at length by hand, which is a narrowing of their capacities, not a modernization of their education.

Hand tools. Civic. Hammer, saw, plane, chisel, file, wrench. Each builds specific physical skills and broader embodied judgment — how materials behave, how force transfers, how tolerances work, how to recover from errors. A person who has spent years with hand tools has an understanding of the physical world that no amount of digital interaction can replicate. The understanding transfers: a carpenter can fix things that aren't buildings, a mechanic can diagnose things that aren't cars, because the habits of inspection and repair apply everywhere.

Cooking from raw ingredients. Civic. A person who cooks develops taste, chemistry intuition, knife skills, timing, improvisation, and the ability to feed themselves and others with whatever is available. These skills survive power outages, supply chain disruptions, and personal financial setbacks. A person who depends entirely on prepared food has none of these capacities; their eating is contingent on the supply chain that delivers their food, and their body is fed by choices other people made about what to include in that food.

Reading long-form text. Civic. Novels, histories, substantial non-fiction. Reading sustained text builds the capacity for sustained attention, complex argument tracking, imaginative reconstruction of scenes not shown, and patience with meaning that accumulates slowly. A person who reads novels regularly becomes someone who can hold structure in mind across hours, which is a capacity needed for almost any serious mental work. A person who has replaced long-form reading with short-form content has lost this capacity, and the loss shows in their ability to engage with any material that doesn't immediately reward them.

Short-form video platforms. Discivic. The platform is designed specifically to fragment attention — each short video resets the user's focus, and the scroll gesture teaches the user to abandon anything that doesn't grab them instantly. Regular use produces shorter attention spans, reduced ability to tolerate delayed gratification, and specific neurological adaptation to constant stimulation. The user spends time on the platform and comes away with nothing that persists — no knowledge, no skill, no artifact, no development. The capacity they brought to the session is slightly smaller when the session ends. Regular use compounds the reduction.

Passive television and streaming consumption. Largely discivic under regular use, with qualifications. Watching a film actively, discussing it, thinking about its construction, revisiting it — this is closer to reading a novel, and can be civic. Leaving the television on for hours as background, scrolling through streaming platforms as a way of filling time, consuming content without engagement — this is substitution for whatever the person would have done with the hours, and the substitution typically produces nothing. The test is whether the person is engaged with the content or being occupied by it.

Driving a manual-transmission car on varied roads. Civic. The driver develops spatial awareness, mechanical intuition, coordination of multiple simultaneous controls, and the ability to read road conditions. The skills transfer to other physical tasks requiring coordination under uncertainty. The driver who has done this for years can handle unfamiliar vehicles and unfamiliar conditions with some confidence because the underlying skills generalize.

Automatic transmissions, lane-keeping assist, adaptive cruise control. Mixed. These technologies are each individually minor, and their civic or discivic character depends on how they're used. A driver who uses lane-keeping as a safety margin while remaining fully attentive is being augmented. A driver who uses it to half-attend while doing other things is being substituted for, and their driving skills erode accordingly.

Fully autonomous vehicles. Discivic by structure for regular users. The person in the car is not driving. They are not developing driving skills. They are not learning the roads in the way a driver learns them. Over years of regular use, a person who has only been driven becomes someone who cannot drive — not because they lack the license, but because they lack the capacity. If the autonomous system fails, they have nothing to fall back on. Children raised in this environment will never develop the skills at all. A civilization that adopts autonomous vehicles universally loses driving as a population capacity within a generation, and with it loses the transferable embodied competence that learning to drive builds.

GPS navigation for routine trips. Discivic when used regularly. The user who follows turn-by-turn instructions on familiar routes loses the mental map they would otherwise develop. Over years, they become someone who cannot navigate without the GPS, even in their own city. Spatial memory and directional sense are real capacities that atrophy without use, and GPS removes the use. A person who navigated by landmarks and paper maps for decades retains the capacity; a person raised on GPS never develops it.

GPS for genuinely unfamiliar travel. Closer to medicine in the food analogy. Used when actually needed, it provides capability the user couldn't easily have on their own — navigating a foreign city they've never visited. Used occasionally and consciously, it doesn't erode the navigation capacity the user has built elsewhere. The civic-discivic character depends on whether the use is situational (when the user is genuinely outside their capacity) or routine (when the user had the capacity and is letting it atrophy).

Social media with algorithmic feeds. Largely discivic. The platform is designed to capture attention, and it does so by triggering emotional responses that keep the user scrolling. Regular use produces measurable changes: shortened attention spans, increased anxiety, reduced in-person social skill, altered self-perception through comparison dynamics, and specific training in passive reception of content. The user ends sessions with no artifact, no skill gained, and subtle degradations in their capacity to function without the platform. The feeds also shape the user's sense of what's important by what the algorithm surfaces, which erodes the capacity for independent judgment about what matters.

Chat-based artificial intelligence used for directed work. Civic for users who stay in the driver's seat. The user asks questions, evaluates answers, directs follow-ups, and integrates what they learn into their own work. Over time, they build knowledge about the subjects they've researched and the skill of directing AI-augmented work. The AI provides capability they couldn't have alone; the direction provides the human capacity-building. This pattern works for adults with prerequisite skills. It does not work the same way for children who lack those skills and cannot maintain the driver's seat.

Agentic artificial intelligence that runs autonomously on user prompts. Largely discivic for typical users. The user submits a task and receives output. They don't engage with the process, don't absorb the subject matter, don't build direction skill because minimal direction is required. The output may be useful; the user ends the session with nothing new in themselves. Regular use of this pattern across many tasks produces someone who can submit prompts and receive outputs, which is a shallow skill. The deep capacities that come from sustained engagement with problems remain undeveloped.

Programming. Civic, despite the physical costs of long sitting. The activity requires the user to develop logical thinking, systematic decomposition, precise language use, debugging skills, and patient engagement with systems. Programmers build both the programs they produce and the mental capacities that produce them. The mental capacities transfer to many non-programming domains — scientific reasoning, careful argument, systems thinking in social and organizational contexts. The physical costs are real but mitigable through posture, movement, and compensating physical activity.

Construction work and skilled trades. Civic. The worker builds physical structures that stand for decades and the embodied skill of someone who can build. The skill transfers across related work and across life — the carpenter can fix things at home, the electrician can diagnose problems in any wiring, the mason understands how solid things go together. The physical costs are real. The trade is honest: body wear for skill, output, and durable capacity that serves the worker and the civilization.

Education with human teachers. Civic when done well. The teacher models thinking, provides feedback, challenges the student, and adjusts to the student's state. The student learns to think, argue, absorb new material, and work through difficulty. These capacities are the product of engaged teaching, not of information transfer. Education reduced to information delivery or automated grading loses the civic character because the teacher-student engagement that built the capacities is absent.

Smartphone-based everything as the child's default interface. Discivic for children, unambiguously. The smartphone substitutes for activities that would have built capacity: reading long-form, writing by hand, playing physically, navigating without the device, waiting without stimulation, interacting face-to-face with other humans. Children who grow up with the smartphone as their default mode of interaction don't develop the capacities that children without smartphones develop. The loss is visible in attention, social skill, physical coordination, and the ability to tolerate the absence of external stimulation.

These examples are not meant to settle debates. Reasonable people will classify some technologies differently for themselves. The purpose is to show how the civic-discivic test is applied: look at what regular use produces in the human, over time, with attention to what's left behind and what's eroded. Specific classifications can vary; the method is the same.

7. Why This Classification Matters

The classification might seem abstract — a vocabulary exercise for evaluating technologies. It matters because decisions about technology are being made constantly, at scales ranging from individual households to civilizations, and the decisions shape the humans who live downstream of them.

At the individual level, knowing whether a technology is civic or discivic for you changes how you should use it. A civic technology can be used regularly without worry — the more you use it, the more you gain. A discivic technology should be used sparingly, or not at all, or in specifically bounded ways that prevent its pattern from becoming your pattern. The classification doesn't tell you to avoid discivic technologies; it tells you to treat them as having costs you need to weigh against their benefits.

At the family level, the classification is decisive for children. Children are in the formation window — the period during which the capacities they'll have as adults are being built by what they engage with. Substituting for formation is different from substituting for adult function. An adult who uses a calculator has already learned arithmetic and the calculator is handling routine cases. A child who uses a calculator before learning arithmetic may never learn arithmetic, because the calculator prevented the practice that would have built the skill. Parents deciding what technologies to put in front of children are deciding what capacities the children will have. The classification gives them a principled basis for the decisions.

At the school level, the same logic applies at scale. A curriculum that substitutes technology for skill-building produces students who haven't built the skills. When those students become adults, they lack the capacities the curriculum replaced. This is why removing cursive, reducing handwritten work, substituting typing for composing, substituting calculators for mental math, and substituting AI-drafted writing for student writing are not neutral choices. Each removal deletes a formation activity and produces adults who lack what the activity would have built. Schools that want to produce capable adults need to preserve formation activities even when the technologies that replace them are more convenient.

At the civilizational level, the aggregate effect of technology choices shapes what the society can do. A society that adopts civic technologies widely and maintains formation activities in its developmental environments produces citizens who are capable of many things. The citizens can work, create, govern, repair, invent, teach, and sustain the civilization they live in. A society that adopts discivic technologies widely, especially in developmental environments, produces citizens who depend on the technologies for their functioning. When the technologies work, the citizens function. When the technologies fail, drift, or become unaffordable, the citizens cannot replace them with their own capacity, because the capacity was never built.

This is the civilizational stake. Civilizations are not permanent. They rise and fall on the capacities of their populations. A civilization of capable humans has reserves to draw on when things go wrong. A civilization of humans dependent on infrastructure has only the infrastructure. When the infrastructure degrades — from war, from economic disruption, from environmental change, from the drift of the technologies themselves toward purposes that don't serve users — the population has no backup. They cannot rebuild, because rebuilding requires capacities they don't have.

The civic-discivic distinction gives civilizations a way to audit their technological choices before the audit becomes necessary through crisis. A society that classifies its technologies honestly and maintains civic activities in its formation environments is investing in the capacities it will need. A society that adopts every technology as equivalent and lets discivic patterns dominate its developmental environments is consuming its future capacity for present convenience.

8. The Generational Test

Within a single generation, the effects of technology are mixed and often hard to see. Current adults who use discivic technologies can still function because they formed before the technologies existed. They have the underlying capacities to fall back on. The technologies erode those capacities over time, but the adults started with enough reserve to absorb the erosion for a while.

Across generations, the effects become unambiguous. Children formed inside discivic technologies don't have the reserve. They never built the capacities the technologies substitute for. Whatever the technologies prevent, the children don't develop. Whatever the technologies replace, the children can't perform without them. By the time those children become adults, the population that was shielded by the reserve of prior-generation skills has been replaced by a population that never had the reserve at all.

This is the generational test for classifying technologies. A technology that works well for current adults and seems harmless may still be discivic if the population formed inside it across generations loses capacities that the current adults kept. A technology's first-generation appearance is not diagnostic. Its third-generation effect is.

The generational test resolves what single-generation observation cannot resolve. Within a lifetime, you see the first-generation effects. Adults who adopt a technology at forty retain most of what they had; the technology may reduce their capacity slightly but their functioning is largely preserved. Their children, adopting the same technology at fifteen, lose more because the adoption intersected with the formation window. Their grandchildren, born into the technology, lose the most because they never had any alternative formation. The full effect is visible only when the generational cascade has completed, and by then the loss is already locked in for the generations that lived through it.

This is why current adults often underestimate the effects of technologies they themselves use without apparent harm. They're protected by the capacity they formed before the technology existed. They judge the technology by its effect on them, and the effect is small. They don't see — can't see, from their position — the effect on children who don't have their reserve. The effect on children is the real classification signal, and it's visible only by looking at the children, not by asking the adults.

The generational test has a further implication. Once a technology has replaced a formation activity for one full generation, the adults who emerge from that generation cannot easily teach the replaced capacity to their children, because they themselves don't have it. The loss compounds: generation one loses the activity but kept the capacity; generation two didn't build the capacity but sees adults around them who have it; generation three has neither the activity nor the adults who can model it. By generation three, the capacity is effectively unrecoverable without deliberate cultural effort against a population that has no reason to value what they never experienced.

The removal of cursive from schools is a current-generation instance of this pattern. The removal happened in a single generation. The adults who made the decision had cursive themselves — they could judge whether it was useful based on their own experience, which was that they didn't use it much in their careers, which became the basis for removal. The children who went through school without cursive don't have the skill. Their children will be taught by teachers who don't have cursive. By the third generation, cursive is gone from the population, recoverable only by specific historical-skills effort. Whatever capacities cursive built — hand-eye coordination, sustained handwritten composition, specific neurological development — are gone with it, and the loss won't be recognizable to a population that never had the capacities.

Every technology adoption in a developmental environment carries this generational risk. The question is not whether current children can use the technology — of course they can, children are adaptable. The question is whether the children, grown, will have the capacities that previous generations had. If yes, the technology is civic for developmental use. If no, the technology is discivic for developmental use, whatever its appeal for adult users with formed capacities.

9. Commercial Incentives and the Problem of Silent Costs

Civic technologies and discivic technologies face different commercial incentives, and the difference explains why the discivic category tends to expand without corresponding growth in the civic category.

Civic technologies typically require active user engagement. The user has to do the work. The technology helps, but the help is useless if the user doesn't bring their effort. This limits how broadly these technologies can scale, because they depend on users who are willing and able to engage. It also limits the revenue they can extract, because user engagement is a limited resource and users don't spend it continuously on any one tool. A hand tool gets put down when the work is done. A book gets finished. A writing session ends.

Discivic technologies, by contrast, are often designed to capture continuous attention or to handle tasks entirely for the user. The user is not required to engage actively; they can receive passively. This design scales well commercially. The user is available as an attention source for as many hours as they'll spend on the platform. The user is willing to delegate tasks as long as the delegation feels like it's saving them work. Revenue scales with use, and use is maximized by making the technology easy to use without effort — which is precisely the design pattern that produces discivic outcomes.

This is not a conspiracy. It's a structural feature of markets for technology products. Products that require effort from users compete with products that don't require effort. Given equivalent short-term benefits, users often choose the effortless version. Providers who build effortless versions capture market share. The market rewards discivic design and penalizes civic design, unless the civic design produces value the user explicitly wants enough to pay for the effort.

This dynamic is amplified by the fact that the costs of discivic technologies are mostly invisible at the point of adoption. When a user adopts a short-form video platform, they don't see their attention span shortening. They feel the platform is entertaining. Only over months and years, and usually only in retrospect if at all, does the attention change become apparent. The cost is real but distributed across time and invisible in any single session. The benefit — entertainment, distraction, convenience — is immediate and visible. The calculation users actually make is biased toward adoption, because the benefit registers and the cost doesn't.

Children are especially vulnerable to this asymmetry. They experience the immediate benefit of a new technology — the stimulation, the ease, the social connection with peers who also use it. They cannot perceive the long-term cost, because the cost lives in capacities they haven't yet built and therefore don't know they should build. A parent deciding what to introduce to their child is making the decision with information the child doesn't have, and the parent's own information is limited by the fact that the effects on children may differ from the effects on adults.

Schools face a version of the same problem. A school adopting a technology for efficiency reasons often can't see what the technology is displacing. The immediate effect is positive — students can produce more, faster, with less teacher time. The long-term effect — that the students haven't developed the skills the old activity built — shows up only when those students become adults, and by then the decision has been normalized and is hard to reverse.

Governments and regulators face an even harder version. The effects of a technology on a population take years or decades to become visible at the population level. By the time the effects are measurable, the technology is deeply embedded in economic and social life, and rolling it back is politically impossible even when the diagnosis is clear. Regulators acting on early evidence face pushback from industries that profit from the technology and from users who have adopted it and don't want to give it up. Regulators acting only after overwhelming evidence find themselves intervening after the damage is done.

This is why the classification has to be done before the full effects are visible. Waiting for conclusive evidence means waiting until it's too late. The test provides a way to evaluate technologies prospectively, based on what they structurally do to users rather than on what the statistical record eventually shows. A technology that substitutes for human capacity in developmental environments is predictably discivic, whether or not the population-level data has accumulated to prove it. The prediction can guide adoption decisions before the evidence would be available, and the evidence, when it arrives, will typically confirm the prediction.

10. The Honest Trade-Off

The paper has argued that technologies should be classified by what they do to humans over time, that civic technologies build capacity and discivic technologies erode it, and that the classification matters for individuals, families, schools, and civilizations. The remaining question is what to do with the classification when your own use of technology is at stake.

The classification does not prescribe. Adults can choose to use discivic technologies for reasons that make sense in their situations. A person working multiple jobs to support their family might use discivic technologies because the time they save is necessary, even knowing the cost. A person with specific disabilities might use technologies that would be discivic for others but are necessary medicine for them. A person who has built substantial capacity through decades of civic activity might use discivic technologies occasionally because their reserve absorbs the cost.

What the classification provides is honest accounting. When you choose a discivic technology, you know what you're choosing. You're trading capacity for something else — convenience, time, productivity, entertainment. The trade might be worth it. The reason to name it as a trade is so that you can decide whether it's worth it rather than treating the cost as zero because it's invisible.

Some users will find that their trade-offs were not worth what they thought. They adopted discivic technologies believing the costs were minor, and over years they notice that their attention is shorter, their patience thinner, their capacity to sustain engagement reduced. The classification lets them reverse course — to identify which technologies are costing them more than they're delivering, and to rebuild the capacities that have eroded.

Some users will find that their trade-offs were worth it, and the classification confirms rather than challenges their choices. They adopted discivic patterns deliberately, got what they wanted from them, and are willing to continue paying the cost. The classification doesn't argue with this. It just makes the accounting visible.

For children, the accounting is different because children can't make the trade-off. They don't have the adult perspective to evaluate costs, don't have the formed capacities to trade against gains, and don't have the agency to accept or reject technologies that adults around them have adopted. Adults who make decisions for children are making the trade-off on the children's behalf, and the classification imposes a specific responsibility: don't let children formatively adopt technologies that would erode capacities they haven't yet built. The adults made the trade-off with their own capacities to trade. The children have no capacity to trade — what they have is the capacity they would have built if given the formation environment that builds it.

Civilizations face the same choice at scale. A civilization can adopt technologies broadly, reap the short-term economic and convenience benefits, and accept the long-term erosion of population capacity as the cost. Some civilizations are making this choice now. The short-term benefits are visible; the long-term cost will show up in a generation or two, by which time the civilization that made the choice will not be the civilization that pays the bill.

Civilizations that want their futures to be capable of more than their presents have to do something different. They have to protect the formation environments from discivic technologies, even when adopting those technologies widely in adult contexts. They have to maintain civic activities in schools, in homes, in public life, even when convenience pressures all of these toward discivic alternatives. They have to build populations whose capacities compound across generations rather than eroding.

This is hard. The pressures run the other way. Every convenience is cheaper than every civic activity. Every piece of content is more engaging than every book. Every automated service is faster than every developed skill. The civic path is the uphill path, and it has to be chosen deliberately, against the constant pull of discivic alternatives that are individually easier and aggregately fatal.

11. What You Can Now Do

The classification is in your hands. You can apply it to the technologies in your own life, look at what each one has been doing to you over time, and decide whether the patterns serve what you want to become. You can apply it to the technologies you put in front of your children, knowing that their formation is at stake and you're making decisions they can't yet make for themselves. You can apply it to the institutions around you — schools, workplaces, public services — and see which ones are building capable humans and which ones are producing dependency.

The vocabulary is yours. Civic technology — it builds the humans who use it, leaves capacity in them that persists whether or not the technology is present. Discivic technology — it substitutes for human capacity, leaves the user less capable than they would have been, or capable only while the technology runs. These terms pick out real features of the world. When you use them to describe technologies, you're describing what the technologies actually do, not what they're marketed as.

The test is transferable. For any technology, ask what remains in the human after regular use. If what remains is more capacity, the technology is civic. If what remains is less capacity, or capacity that only works while the tool is running, the technology is discivic. Apply the test to cases you've been uncertain about. Apply it to cases others argue for or against. The test will sort them.

The generational frame is available. Within a lifetime, effects are mixed and often hidden. Across generations, they become clear. A technology that seems fine for current adults may be deeply harmful for children raised inside it. When you're evaluating a technology's deployment in formation environments, use the generational frame. Ask what the third generation of users will be like. The answer is more diagnostic than anything the first generation can report about itself.

The stakes are understood. This is not a minor matter of preference. The technologies a civilization adopts shape what its people can do. Civic technologies compound across generations into more capable populations. Discivic technologies compound across generations into less capable populations that depend on their infrastructure to function. The direction a civilization goes on this question determines what it will be able to do in the future, and the decisions that determine the direction are being made now, often without this frame being applied.

You are not responsible for the whole civilization. You are responsible for your own technology choices, for the technologies you introduce to children in your care, and for the arguments you make to others about these questions. With the classification, the vocabulary, the test, and the generational frame, you can make these choices and arguments on honest terms. Other people will make different choices. The civilization-level outcome is the aggregate of individual choices made with whatever frames individuals hold. Adding this frame to more people's thinking changes the aggregate in the direction of populations that remain capable of the things that matter.

12. Closing

Every technology is a proposition about what its users will become. Some propositions expand the humans who accept them. Others diminish the humans who accept them. The propositions are being made continuously, by every technology adopted into a life or a school or a society. The humans accepting them are being shaped accordingly, whether or not they notice.

Civilizations are not maintained by their technologies. They are maintained by the capacities of their people, which the technologies can build or erode. A civilization of capable people can survive the failure of most technologies; a civilization of dependent people cannot survive the failure of any technology its dependency rests on. The technologies come and go. The people, and what they can do, are what persists — or fails to persist — across the generations.

The distinction in this paper is meant to be used. Classify the technologies in your life. Classify the ones you're putting in front of children. Classify the ones your schools and workplaces have adopted. Some will be civic; use them freely. Some will be discivic; use them with awareness of the cost, or not at all, depending on what you can afford to lose. Help the people around you make the same classifications. Argue for formation environments that protect the building of human capacity, against the convenience pressures that would substitute for it.

You live inside a civilization whose technological choices are being made largely without this frame. You can add it to your own choices. You can contribute it to the discussions you're part of. You can build the next generation of humans in your care with attention to what the technologies are doing to them, rather than treating all technologies as equivalent delivery mechanisms for their advertised features.

The distinction is simple. Civic technology expands humans. Discivic technology reduces them. Most technologies are one or the other. Some can be either, depending on how they're used, and careful use can shift mixed cases toward civic outcomes. The classification is yours to apply, and the classifications you make will shape the futures of the people you affect, starting with yourself and extending to whoever you're responsible for.

The future is not built by technologies. It's built by the humans the technologies produce. Choose accordingly.

References

[**HOWL-SOPH-3-2026**] Civic and Discivic Technologies. Github: https://github.com/ghowland/papers/tree/main/papers/S
SOPH-3-2026

SOPH-3 Supporting Appendix Tables

Appendix A — The Civic/Discivic Classification at a Glance

Legend for Status column:

- **C** — Civic: builds user capacity over regular use
- **D** — Discivic: erodes user capacity over regular use
- **M** — Mixed: depends on how the user engages; can go either direction
- **S** — Situational: functions as medicine in specific conditions; neither civic nor discivic in isolation

Status	Principle	What Remains in the User
C	Requires active engagement; builds skill through use	More capacity than before; persists without the tool
D	Substitutes for human activity; captures attention without building	Less capacity than before; function depends on tool
M	Neutral artifact, outcome determined by user's position	Civic if user stays in driver's seat, discivic if user delegates
S	Intervention in situations where capacity isn't available	No lasting effect on capacity at appropriate use frequency

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Appendix B — Communication Technologies

Technology	Status	Rationale	Civic Mode	Discivic Mode
Spoken language	C	Core human coordination; develops through use	Conversation, argument, teaching	—
Written language	C	Thought externalization; builds composition capacity	Composing from own mind	—
Handwriting (block)	C	Fine motor, hand-eye, memory encoding	Note-taking by hand	—
Handwriting (cursive)	C	Sustained handwritten thought, specific motor/neural dev	Long-form writing, journals	—
Typing	M	Fast transcription of already-formed thought	Typing own composed text	Typing as substitute for handwriting practice in formation

Technology	Status	Rationale	Civic Mode	Discivic Mode
Reading long-form text	C	Sustained attention, structure-holding, mind-modeling	Novels, histories, substantial non-fiction	—
Reading short-form content	D	Trains attention fragmentation when it's the default mode	—	Primary reading diet of headlines, posts, snippets
Letter writing	C	Sustained composition to a specific person	Hand-written or typed letters of substance	—
Voice telephone calls	C	Real-time verbal exchange, tone reading, improvisation	Conversation with persons	—
Texting (SMS / chat)	M	Short written exchange	Used with thought-before-sending	Used as continuous dopamine-triggered reaction
Emoji-heavy messaging	D	Substitutes emotional signal for worded expression	—	Default mode of response
Email (drafted yourself)	C	Long-form asynchronous written exchange	Composed from own thought	—
Email (AI-drafted)	D	Production is by LLM; user receives draft	—	Accepting drafts with minimal revision
Voice messages	M	Asynchronous spoken content	Thoughtful recording	Default mode replacing text and calls
Video calls	M	Face-to-face at distance	Meaningful conversation	Background/passive use

Technology	Status	Rationale	Civic Mode	Discivic Mode
Social media posting	M	Personal broadcast	Sharing substantive work/thought	Performative posting for engagement
Algorithmic social media feeds	D	Attention capture with engagement-optimized content	—	Any regular use
Short-form video platforms	D	Structural attention fragmentation by design	—	Any regular use

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Appendix C — Writing and Composition Technologies

Technology	Status	Rationale	Civic Mode	Discivic Mode
Pen and paper	C	No interference; full writer agency	All composition	—
Pencil	C	Same as pen, plus erasure for revision	All composition	—
Typewriter	C	Linear typing, no correction layer	Composing at keyboard	—
Text editor (vi, emacs, basic)	C	Clean text entry, user composes	Writing code, documents	—
Word processor with spell-check	M	Surface correction of your text	Used after composition	Used as writing crutch before spelling is learned
Grammar/style checker	M	Suggestions on your text	Reviewing as second-opinion	Accepting all corrections by default
Autocomplete while typing	D	Tool proposes ahead of user's thought	—	Any regular use; structurally inhibits formation

Technology	Status	Rationale	Civic Mode	Discivic Mode
Smart compose (sentence prediction)	D	Tool generates sentences user accepts	—	Any regular use
AI-drafted writing (chat, user edits)	M	User prompts, LLM drafts, user revises	Used for research/formatting of own ideas	Accepting drafts as final, especially in formation
Agentic AI writing	D	Task submitted, output received	—	User not engaged with writing process
Dictation software	M	Voice-to-text conversion	For users who compose verbally	Replacing writing practice in formation
Blank page (any medium)	C	Formation condition for original thought	Facing it and producing	—

Appendix D — Physical Tools and Embodied Work

Technology	Status	Rationale	Civic Mode	Discivic Mode
Hammer	C	Force transfer; builds precision and judgment	Any use	—
Saw (hand)	C	Cutting skill, line-following, patience	Any use	—
Plane, chisel, file	C	Fine material work, tolerance development	Any use	—
Power tools	M	Augment hand skill with force/speed	Used by someone with hand-tool base	Used as substitute for learning hand-tool work

Technology	Status	Rationale	Civic Mode	Discivic Mode
Sewing by hand	C	Fine motor, material understanding	Any use	—
Sewing machine	M	Production speed with hand skill base	Used with sewing knowledge	Used without hand-sewing skill
Knitting, weaving	C	Pattern-holding, motor skill, material feel	Any use	—
Woodworking (traditional)	C	Embodied judgment, material knowledge	Any use	—
Masonry	C	Structural understanding, material behavior	Any use	—
Metalwork / smithing	C	Material science through direct work	Any use	—
Gardening (hands-on)	C	Plant knowledge, soil understanding, patience	Any use	—
Cooking from raw ingredients	C	Taste, chemistry, knife skills, feeding yourself	Any use	—
Pre-cut, pre-packaged cooking	D	Removes skill-building steps	—	As primary cooking mode
Food delivery (as primary)	D	No cooking practice at all	—	As primary food source
Hunting, fishing, foraging	C	Ecological knowledge, patience, direct food sourcing	Any use	—

Technology	Status	Rationale	Civic Mode	Discivic Mode
Musical instruments	C	Motor skill, ear training, sustained practice	Any use	—
Drawing, painting, sculpture	C	Visual perception, hand-eye, creative production	Any use	—
Physical sports	C	Coordination, strategy, embodied judgment	Any participation	—
Weight training / calisthenics	C	Body capacity, kinesthetic sense	Any use	—
Walking, hiking, cycling	C	Spatial sense, cardiovascular, attention	Any use	—

Appendix E — Computation and Development Tools

Technology	Status	Rationale	Civic Mode	Discivic Mode
Terminal / CLI	C	Predictable tool; user specifies, tool executes	Any use	—
Text editor (vi/emacs)	C	User-driven text manipulation	Any use	—
Compilers	C	Deterministic code translation	Any use	—
Version control (git, used directly)	C	Project state management	Any use	—
Debuggers	C	Systematic investigation; builds diagnostic capacity	Active debugging	—

Technology	Status	Rationale	Civic Mode	Discivic Mode
Programming (any language)	C	Logical decomposition, precise language, problem-solving	Any serious engagement	—
Mathematical notation / proofs by hand	C	Structured reasoning, precision	Any use	—
Calculator (after arithmetic fluency)	M	Speeds arithmetic for someone who has it	Use alongside mental math	Use before arithmetic is learned
Computer algebra systems	M	Symbolic computation	Used to verify own derivation	Used as black box in formation
Spreadsheets	M	Data structuring and computation	User builds formulas	Accepting templates without understanding
Databases (hand-designed queries)	C	Structured data reasoning	Any use	—
Chat-based AI for directed research	M	Knowledge augmentation with user in driver's seat	User asks, evaluates, directs	User delegates, accepts uncritically
AI code assistants (suggestions on request)	M	Reference lookup during coding	Requested for specific questions	Accepting all suggestions as primary authorship
AI code assistants (continuous autocomplete)	D	Code appears ahead of the programmer's thought	—	Default-on during learning or regular work
Agentic AI (autonomous task completion)	D	User prompts, receives completed work	—	Any regular use for tasks user could do
Integrated development environments (IDE)	M	Augmentation of basic editor	Used with understanding of what it's doing	Used as substitute for understanding

Technology	Status	Rationale	Civic Mode	Discivic Mode
Cloud services as abstraction layers	M	Infrastructure you don't run	Understanding what's happening underneath	Total black-box dependency

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Appendix F — Navigation and Transportation

Technology	Status	Rationale	Civic Mode	Discivic Mode
Navigation by landmarks	C	Builds spatial memory, observational skill	Any use	—
Navigation by stars/sun	C	Deep ecological and astronomical knowledge	Any use	—
Paper maps	C	Builds mental map, scale sense, geography	Any use	—
Compass	C	Orientation without the map; embodied direction	Any use	—
Walking as primary transport	C	Full spatial awareness, body capacity	Any use	—
Bicycling	C	Balance, distance sense, embodied navigation	Any use	—
Driving (manual transmission)	C	Coordination, mechanical intuition, attention	Any use	—
Driving (automatic transmission)	M	Loses gear-shifting skill	Still requires road judgment	Paired with further automation, full substitution

Technology	Status	Rationale	Civic Mode	Discivic Mode
Driver-assist features (adaptive cruise, lane-keep)	M	Safety margins vs. attention substitution	Used as backup while fully engaged	Used to half-attend while driving
GPS for genuinely unfamiliar travel	S	Extends capacity into new territory	Occasional use when needed	—
GPS for routine/familiar trips	D	Substitutes for mental map you'd otherwise build	—	Default use in known areas
Turn-by-turn voice navigation	D	User follows instructions rather than navigating	—	Default mode of getting anywhere
Autonomous vehicles	D	User does not drive; no skill develops or is used	—	Any regular use
Public transit (with route reading)	M	Some navigation, some reliance	Reading routes, understanding geography	Passive following of app instructions

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Appendix G — Learning and Education Technologies

Technology	Status	Rationale	Civic Mode	Discivic Mode
Human teachers with engaged students	C	Modeling thinking, feedback, adaptation	Any engaged teaching	—
Apprenticeship	C	Master-to-student skill transfer over time	Any serious engagement	—

Technology	Status	Rationale	Civic Mode	Discivic Mode
Classroom instruction (well-done)	C	Shared learning, question-driven exchange	Engaged students	—
Classroom instruction (lecture-only, unengaged)	M	Information transfer without thinking development	If student does the thinking independently	If student just absorbs and restates
Textbooks	C	Structured exposition for self-directed study	Working through with pencil/paper	—
Libraries / archives	C	Access to accumulated knowledge	Using them for inquiry	—
Flash cards and memorization tools	M	Repetition for retention	Supporting understanding	Memorization without comprehension
Educational videos (active watching)	M	Visual explanation	Paired with practice and note-taking	Passive watching as primary mode
Khan-academy-style structured learning	M	Self-paced instruction	Working problems alongside	Watching without doing
AI tutors (Socratic-style, user in driver's seat)	M	Adaptive questioning assistant	Student does the thinking	Student accepts AI reasoning as their own
AI tutors (solution-providers)	D	Answer delivery rather than teaching process	—	Any regular use in formation
Homework done by AI	D	No learning happens for the student	—	Any regular use
Note-taking by hand	C	Encoding through the hand, selective attention	Any use	—

Technology	Status	Rationale	Civic Mode	Discivic Mode
Note-taking by typing	M	Faster capture, less encoding	Active re-processing later	Verbatim transcription without processing
AI-generated notes/summaries	D	Student bypasses processing entirely	—	Primary mode of note creation
Formal math by hand	C	Deep reasoning practice	Any use	—
Laboratory work (hands-on)	C	Direct engagement with phenomena	Any use	—
Simulated labs (virtual)	M	Exposure to concept	Paired with real labs	Replacing real labs in formation

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Appendix H — Entertainment and Leisure

Technology	Status	Rationale	Civic Mode	Discivic Mode
Novels and long-form fiction	C	Sustained attention, mind-modeling, imagination	Any regular reading	—
Poetry	C	Language density, rhythm, meaning compression	Reading or writing	—
Non-fiction (substantial)	C	Idea absorption, structured argument	Any regular reading	—
Short stories	C	Narrative craft, focused attention	Any regular reading	—

Technology	Status	Rationale	Civic Mode	Discivic Mode
Magazines (substantive)	M	Variable quality; surface or deep	Engaged reading of substance	Idle flipping through ads
Newspapers (physical)	M	Current events with sustained reading	Engaged reading of substance	Scanning headlines only
Online news (sites, not feeds)	M	Current events, variable depth	Reading full articles by choice	Skimming headlines algorithmically served
Blogs (substantive)	C	Long-form argument and exposition	Regular engaged reading	—
Podcasts (long-form, focused)	M	Audio long-form	Active listening, thinking	Background noise while scrolling
Films (watched actively)	C	Narrative absorption, visual literacy	Full attention to quality work	—
Streaming series (thoughtful engagement)	M	Long-form visual narrative	Deliberate viewing of substance	Binge-consumption as attention fill
Streaming (auto-play background)	D	Passive consumption without engagement	—	Default mode
Radio (talk, music)	M	Variable engagement	Active listening	Passive background
Music (active listening)	C	Attention to craft, emotional range	Listening to work deliberately	—
Music (background while other activity)	M	Ambient exposure	Appropriate to the activity	Constant stimulation avoidance
Video games (complex, skill-building)	C	Strategic thinking, coordination, narrative	Games that demand engagement	—
Video games (reflex-only, loot-box)	M	Variable; some build reflexes, some extract	Skill-building engagement	Compulsion-loop design

Technology	Status	Rationale	Civic Mode	Discivic Mode
Mobile games (casual)	D	Usually attention-capture without skill	—	Any regular use
Gambling apps	D	Attention and financial capture	—	Any use
Board games, card games	C	Strategy, social interaction, rule-holding	Any regular play	—
Chess, Go	C	Deep strategic thinking	Any regular play	—
Puzzles (crossword, sudoku, jigsaw)	C	Focused problem-solving	Any regular practice	—
Pornography (general access)	D	Reward-circuit hijacking, relationship distortion	—	Regular use
Short-form video platforms	D	Structural attention fragmentation	—	Any regular use
Algorithmic social feeds	D	Comparison dynamics, attention capture	—	Any regular use
In-person social gatherings	C	Social skill, conversation, shared experience	Any participation	—
Outdoor activities	C	Physical capacity, environmental awareness	Any engagement	—

Appendix I — AI and LLM Technologies (Detailed)

Technology	Status	Rationale	Civic Mode	Discivic Mode
Chat interface for factual lookup	M	Reference tool	Occasional questions, verified answers	Default epistemic authority without verification
Chat interface for directed research (user-led)	C	Knowledge augmentation with driver's-seat user	User asks, evaluates, directs, absorbs	—
Chat interface used for delegation	D	Output received without engagement	—	"Do this for me" patterns
Chat interface for drafting with user revision	M	Collaborative production	User composes, LLM refines at user's direction	User accepts drafts as final
Chat for therapy / emotional support	M	Conversational companion	Processing one's own thoughts with reflection	Substituting for human relationship or real help
Chat as writing tutor	M	Adaptive feedback	User composes, gets targeted feedback	User outsources composition entirely
Chat as programming helper (asked questions)	M	Reference during coding	Directed lookup of specific issues	Continuous delegation of problem-solving
Inline autocomplete in any writing	D	Suggestions appear ahead of thought	—	Any regular use, especially in formation
AI email drafting	D	User receives drafts to send	—	Default mode of email composition
AI summarization of text user hasn't read	D	User bypasses the reading	—	Any regular use
AI translation (direct, no user involvement)	M	Language bridge	Used to access texts otherwise unreadable	Replacing language learning

Technology	Status	Rationale	Civic Mode	Discivic Mode
AI image generation	M	Creative tool	Used alongside drawing/design practice	Replacing art-making capacity
AI music generation	M	Composition tool	Used alongside musical practice	Replacing musical capacity
AI code generation (autonomous)	D	Code produced, user accepts	—	Any regular use in formation or for routine work
Agentic AI (task completion)	D	User prompts, receives result	—	Any regular use for tasks user could do
AI decision-making (recommendations as defaults)	D	Algorithmic choice replacing user judgment	—	Accepting recommendations passively
AI-mediated social interaction	D	Relationship mediated through AI interpretation	—	Regular use
AI companions / partners (romantic, social)	D	Substitutes for human relationship skill-building	—	Any serious use
Voice assistants for simple queries	M	Convenient lookup	For actual hands-busy cases	Default mode over self-resolution
AI for creative brainstorming (with user filtering)	M	Ideation partner	User generates alongside, filters, develops	User accepts AI output as their creativity
AI-generated study materials	D	Student receives rather than constructs	—	Any regular use in formation

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Appendix J — Interface and Input Technologies

Technology	Status	Rationale	Civic Mode	Discivic Mode
Pen/pencil on paper	C	Rich tactile feedback, fine motor, embodied	Any use	—
Keyboard (physical, tactile)	C	Fine motor, muscle memory, feedback	Any use	—
Mouse	C	Hand-eye coordination	Any use	—
Touchpad	M	Less feedback than mouse	Functional	If replacing more skilled input entirely
Touchscreen (tapping)	D	Uniform glass, no material feedback, repetitive strain	—	As primary/only input device
Stylus on screen	M	Approximates pen with digital	Used for drawing/writing practice	If replacing pen/paper in formation
Voice input	M	Different modality	For accessibility or hands-busy cases	Replacing handwriting/typing in formation
Gesture control	D	Repetitive motion without feedback	—	Primary input mode
Eye-tracking interfaces	S	Accessibility medicine	For those who need it	Not yet a mass input
Brain-computer interfaces	S	Emerging; unknown profile	Not yet applicable	Not yet applicable
Physical keyboards for typing	C	Tactile feedback, finger strength, muscle memory	Any use	—
On-screen keyboards	M	Necessary on touch devices	Occasional	As primary typing surface in formation

Appendix K — Home, Family, and Daily Life

Technology	Status	Rationale	Civic Mode	Discivic Mode
Home cooking	C	Feeding capacity, family connection	Any regular practice	—
Home repair and maintenance	C	Practical skill, house knowledge	Any engagement	—
Gardening at home	C	Food and plant knowledge	Any practice	—
Raising children (hands-on)	C	Deep caregiving capacity	Active parenting	—
Smart home automation (full)	D	Removes daily physical engagement	—	As primary home interaction
Smart home (limited, thoughtful)	M	Some convenience	Used with awareness	Full surrender to automation
Household cleaning by hand	C	Physical engagement, attention to space	Any practice	—
Robotic cleaning devices	M	Time savings	Paired with some hand cleaning	Complete substitution
Washing clothes (with machine)	M	Basic modern technology	Standard use	—
Hand-washing dishes	C	Sensory engagement, meditative	Any practice	—
Dishwasher	M	Basic convenience	Standard use	—
Handwriting thank-you notes / letters	C	Personal connection, sustained attention	Any practice	—
In-person conversation at meals	C	Social skill, family bonds	Any practice	—

Technology	Status	Rationale	Civic Mode	Discivic Mode
Screens at meals	D	Replaces conversation	—	Regular pattern
Bedtime reading (to children, by children)	C	Literacy foundation, attention, bonding	Any practice	—
Children's screen time (formation years)	D	Substitutes for formative activity	—	Any regular use in pre-formation window
Physical play (outdoor, unstructured)	C	Body, creativity, social negotiation	Any practice	—
Board games as family activity	C	Strategy, social skill, sustained attention	Any practice	—

Appendix L — The Driver's Seat Test for Mixed Technologies

For technologies with status M (mixed), this test determines civic vs. discivic function for the specific user and pattern.

Signal of Driver's Seat (Civic Mode)	Signal of Passenger Seat (Discivic Mode)
User forms the question before seeking the tool	Tool surfaces questions for user to react to
User evaluates tool output against own judgment	User accepts tool output as authoritative
User verifies with other sources	User treats tool as single source of truth
User could do the task (slower) without the tool	User cannot do the task at all without the tool
User becomes more capable over time	User becomes less capable over time
User directs the process at each step	Process runs with minimal user input
User absorbs subject matter through direction	User receives output without absorbing content
User retains ability to recognize bad output	User has no basis for recognizing bad output
Engagement is active throughout session	Engagement is passive after initial request
Tool is transparent about what it's doing	Tool is opaque; user sees only results

Appendix M — The Generational Test

For any technology being introduced into developmental environments (homes with children, schools, childcare), apply the generational test.

Question	If Yes	If No
Do current adults have a capacity this technology substitutes for?	Check: was the capacity built before the technology existed?	Technology is new enough that the question is prospective
Does using the technology prevent children from building that capacity?	Discivic for formation; protect the formation activity	Possibly civic in formation if capacity forms alongside
Will adults raised on this technology be able to teach the capacity to their children?	Capacity can persist if intentional effort is made	Capacity will be lost by the grandchild generation
Does the technology require prerequisite skills to use in civic mode?	Children cannot use it in civic mode — discivic for formation	Children can potentially use it in civic mode
Is the capacity being substituted for load-bearing (useful across many domains)?	Loss is serious; formation activity must be protected	Loss may be acceptable if gains are real
Would the civilization be in trouble if this capacity disappeared from the population?	Capacity must be preserved; technology should not dominate formation	Capacity loss is tolerable if other capacities are built
Is the technology structurally optimized to capture attention/substitute for activity?	Likely discivic at scale regardless of individual use	Likely civic or mixed

Appendix N — Failure Modes in Applying the Classification

Common errors when using the civic/discivic distinction, and how to recognize them.

Error	Description	Correction
Judging by output	Assuming impressive output means the user is being built	Ask what's in the user, not in the artifact

Error	Description	Correction
Judging by immediate experience	"I feel more productive" as proof of civic use	Check six-month and one-year trajectories of underlying capacity
Generalizing from current adults	Assuming effects on adults match effects on children	Run generational test separately
Treating convenience as benefit	Counting time saved as net gain	Count what wasn't built in the time saved
Treating every tech as neutral	"It's just a tool, depends on how you use it" applied to all cases	Some technologies are structurally discivic regardless of use
Confusing medicine with food	Using situational aids as daily routines	Ask: am I using this because I need it now, or because it's always there?
Ignoring design defaults	Focusing on theoretically possible use, not typical use	Aggregate effects follow the default, not the optimized use
Nostalgia fallacy	Assuming older technologies are automatically civic	Apply the test honestly to old tools too
Novelty fallacy	Assuming newer technologies are automatically discivic	Apply the test honestly to new tools too
Individual-scale blindness	Assuming what's fine for you is fine at scale	Population effects differ from individual effects, especially in formation
First-generation blindness	Using first-generation effects as final verdict	Effects compound across generations; await the generational test
Binary classification pressure	Forcing mixed technologies into C or D	Most real technologies are mixed; document the conditions

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Appendix O — Ancient Technologies Audited

A sample of long-established technologies, showing the classification is not a function of age.

Technology	Era Introduced	Status	Civic Mechanism
Spoken language	Prehistoric	C	Builds coordination, thought, transmission
Fire use	Prehistoric	C	Builds knowledge of material, food preparation, craft
Stone tools	Prehistoric	C	Builds precision, material understanding
Agriculture	~10,000 BCE	C	Builds ecological knowledge, patience, planning
Writing systems	~3,500 BCE	C	Builds thought externalization, record, memory
Wheel	~3,500 BCE	C	Builds engineering intuition, extends transport without substituting for walking
Metallurgy	~6,000 BCE	C	Builds material science, craft, chemistry
Mathematics (notation)	Ancient	C	Builds abstract reasoning
Musical notation	Medieval	C	Builds ear, transmits composition
Printing press	1440s CE	C	Amplifies writing without substituting for it
Sailing	Ancient	C	Builds navigation, ecological knowledge
Carpentry tools	Ancient	C	Builds embodied material knowledge
Masonry tools	Ancient	C	Builds structural understanding
Classrooms	Ancient	C (when engaged)	Builds shared learning
Libraries	Ancient	C	Builds access to accumulated knowledge

Technology	Era Introduced	Status	Civic Mechanism
Weaving, pottery	Ancient	C	Builds embodied craft
Optics (early lenses)	13th century CE	C	Extends sight without replacing it
Mechanical clocks	14th century CE	M	Time coordination; replaces some natural time sense
Firearms	14th century CE	M	Combat capacity; can replace more skilled combat forms
Printing of maps	Renaissance	C	Amplifies geography; paired with travel, builds spatial knowledge
Telescope, microscope	17th century CE	C	Extends observation; builds scientific capacity
Steam engine	18th century CE	M	Amplifies work; can replace physical labor capacity
Telegraph	19th century CE	M	Communication speed; doesn't replace writing or conversation
Telephone	19th century CE	M	Voice at distance; preserves conversation skill
Photography	19th century CE	M	Visual record; can replace drawing
Automobile	20th century CE	M	Transport; varies by user engagement
Radio	20th century CE	M	Broadcasting; varies by listener engagement
Television	20th century CE	M	Visual broadcast; passive mode is discivic

Appendix P — Applying the Classification: Quick Reference Card

For anyone applying the test in practice.

Step 1: Identify the technology and the pattern of use.

Not just "do I use X," but "how often, in what context, replacing what activity."

Step 2: Ask what regular use produces in the human.

More capacity? Same capacity? Less capacity? Capacity only while the tool runs?

Step 3: Check the driver's seat for mixed technologies.

Am I directing, evaluating, and absorbing? Or receiving, accepting, and delegating?

Step 4: Apply the generational test if formation is involved.

Is this going into a child's formation environment? Will the capacity survive in the grandchild generation?

Step 5: Classify.

Civic: use freely.

Discivic: use with awareness of the cost, in bounded ways, or not at all.

Mixed: use in civic mode; structure your engagement to protect the driver's seat.

Situational: use when needed; don't let it become routine.

Step 6: Decide.

Your decisions about technology use are yours. The classification gives you the frame for honest choice. With honest framing, you may still choose a discivic technology — but you'll know the trade you're making, and you'll know what you're paying for what you're getting.

End of Supporting Appendix Tables.